

Blessed P. Nyathi

Software Engineering Intern

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PROFESSIONAL SUMMARY

Second-year Software Engineering student at Tomsk State University with a strong foundation in mathematics, statistics, and programming. Building practical experience through projects spanning cybersecurity, simulation modelling, and machine learning. Seeking a software engineering internship to apply and deepen technical skills in a professional environment.

EDUCATION

BSc Software Engineering

National Research Tomsk State University

Sep 2024 – Aug 2028

Tomsk, Russia

- Core courses: Programming Languages, Linear Algebra, Calculus, Simulation Modelling, Machine Learning, Cybersecurity, etc.
- Zimbabwe Presidential Scholarship recipient.

ZIMSEC A-Levels — 19 Points

Fletcher High School

Jan 2019 – Jan 2021

Gweru, Zimbabwe

- A in Mathematics, Statistics, Chemistry; B in Physics.
- Best Physics Student award (Form 5).

PROJECTS

Secure Online Banking Platform

Cybersecurity Course Project (Team of 3) — TSU

Feb 2026 – Present

- Designed a 7-table PostgreSQL schema implementing double-entry bookkeeping, RBAC roles, per-account transaction limits, and an immutable audit log; hosted on Neon free tier.
- Applied the STRIDE threat model to systematically identify and mitigate spoofing, tampering, repudiation, information disclosure, denial-of-service, and privilege escalation threats.
- Architected a three-tier system: Python Flask frontend, C#/.NET Core backend API with JWT authentication, and PostgreSQL database.
- Mapped security controls to ISO/IEC 27001:2022 Annex A and performed a gap analysis.

SKILLS: PostgreSQL, C#/.NET Core, Flask, JWT Authentication, STRIDE, ISO 27001, Database Design, REST APIs, RBAC

LithiSim-Alpha — Mining Transport Simulator

Personal Project — [GitHub](#)

May 2026 – Present

C++ / Python

- Built a discrete-event simulation engine in C++ modelling truck dispatch, queuing, and stochastic breakdowns in a mining transport scenario.
- Implemented a reusable stochastic model hub with Poisson, exponential, and uniform-integer distributions for dispatch scheduling and travel-time sampling.
- Designed a truck state machine (idle → hauling → dumping → broken) with health-aware dispatch policy and configurable parameters.
- Structured the project with CMake for cross-platform builds; planning a Python analysis layer for log ingestion and policy comparison.

SKILLS: C++17, CMake, Object-Oriented Design, Stochastic Modelling, State Machines, Simulation Design

- **Task 1 — Automated Support-Ticket Processor:** Built a multi-stage LLM pipeline with prompt chaining (preprocess → classify → respond), routing into 4 categories, parallel sentiment analysis & keyword extraction, and a reflection loop for self-improving responses.
- **Task 2 — Domain-Routed Research Assistant:** Developed a pipeline with query decomposition, domain-specific route handling, best-of-N candidate generation with judge selection, fan-in synthesis, and a reflexion-style producer-critic improvement loop with plateau detection.
- Integrated OpenRouter LLM APIs with retry/timeout/fallback resilience, structured JSON rubrics, and unified-diff logging across reflection iterations.

SKILLS: LLM APIs, Prompt Engineering, Agentic Patterns, Python, Async Parallelization, Reflection/Critique

EXPERIENCE

STEM Educator (A/O Level)

Oasis College

Jun 2022 – Jul 2023
Beitbridge, Zimbabwe

- Taught Mathematics, Physics, and Biology to Cambridge and ZIMSEC curriculum classes.
- Enhanced student outcomes by 20% through data-driven instruction and personalized learning plans.
- Designed lesson plans incorporating technology and hands-on experiments.

Projects Director, Interact Club

Fletcher High School

Jul 2020 – Jan 2021
Gweru, Zimbabwe

- Raised \$500 over 12 months through vendor partnerships and fundraising campaigns.
- Coordinated community service projects and led a team of student volunteers.

VOLUNTEERING

Student Volunteer

TSU Online

Oct 2024 – Present
Tomsk, Russia

- Assist international students with administrative processes, immigration, and English–Russian translation.

TECHNICAL SKILLS

Proficient: Python, C++, R, Excel, Mathematics, Statistics
Learning: C#/.NET Core, Flask, PostgreSQL, HTML/CSS, Git, REST APIs, LLM Prompt Engineering, CMake
Concepts: STRIDE Threat Modelling, RBAC, JWT Auth, Database Schema Design, Prompt Chaining, Discrete-Event Simulation

LANGUAGES

English (Proficient) · Ndebele (Native) · Russian (Intermediate) · Shona (Proficient) · Venda (Proficient)

ACHIEVEMENTS

- **Zimbabwe Presidential Scholarship** (2022/2023) — Full university scholarship for academic excellence.
- **Best Physics Student Award** (Form 5) — Fletcher High School.
- **Senior Prefect** — Fletcher High School leadership role.